



| Category  | Item                | Value | Unit |
|-----------|---------------------|-------|------|
| Material  | Concrete            | 100   | m³   |
|           | Steel               | 50    | kg   |
|           | Brick               | 200   | kg   |
|           | Wood                | 30    | m³   |
| Labor     | Construction Worker | 10    | hr   |
|           | Electrician         | 5     | hr   |
|           | Plumber             | 3     | hr   |
|           | Painter             | 2     | hr   |
| Equipment | Excavator           | 1     | hr   |
|           | Truck               | 2     | hr   |
|           | Generator           | 1     | hr   |
|           | Drill               | 1     | hr   |

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## 9702/52

October/November 2010

**1 hour 15 minutes**

No Additional Materials are required.

Write your Centre number, candidate number and name on all the work you hand in.  
Write in dark blue or black pen.  
You may use a soft pencil for any diagrams, graphs or rough working.  
Do not use staples, paper clips, highlighters, glue or correction fluid.  
**DO NOT WRITE IN ANY BARCODES.**

Answer **all** questions.

You may lose marks if you do not show your working or if you do not use appropriate units.

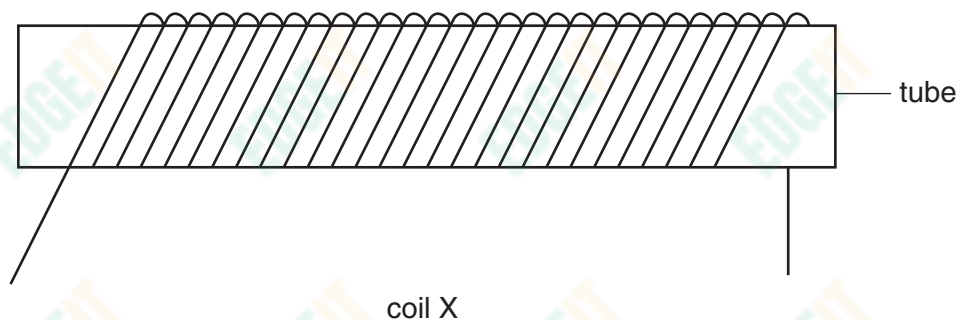
At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use |  |
|--------------------|--|
| 1                  |  |
| 2                  |  |
| Total              |  |

This document consists of **8** printed pages.

- 1 Fig. 1.1 shows a coil (coil X).



**Fig. 1.1**

A student winds another coil (coil Y) tightly around coil X.

A changing e.m.f. in coil X induces an e.m.f. in coil Y.

The student wishes to investigate how the e.m.f.  $V$  in coil Y depends on the frequency  $f$  of the current in coil X.

It is suggested that  $V$  is directly proportional to  $f$ .

Design a laboratory experiment to investigate the suggested relationship. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

- (a) the procedure to be followed,
- (b) the measurements to be taken,
- (c) the control of variables,
- (d) the analysis of the data,
- (e) the safety precautions to be taken.

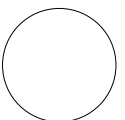
[15]

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**[Turn over**

This image shows a full page of blank handwriting practice paper. It features multiple sets of horizontal lines. Each set consists of a solid top line, a dashed middle line, and a solid bottom line, providing a guide for letter height and placement. The background is white, and the lines are printed in a light gray or blue color. There is no text or other markings on the page.

| For Examiner's Use | Defining the problem | Methods of data collection | Method of analysis | Safety considerations | Additional detail |
|--------------------|----------------------|----------------------------|--------------------|-----------------------|-------------------|
|                    |                      |                            |                    |                       |                   |



- 2 A student is investigating how the period  $T$  of a simple pendulum depends on its length  $l$ , as shown in Fig. 2.1.

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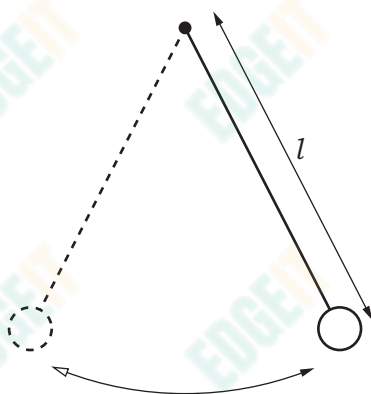


Fig. 2.1

The time  $t$  for 10 oscillations is recorded for a pendulum of length  $l$ . The period  $T$  of the pendulum is determined. The procedure is then repeated for different lengths.

Question 2 continues on the next page.

It is suggested that  $T$  and  $l$  are related by the equation

$$T = al^b$$

where  $a$  and  $b$  are constants.

- (a) A graph is plotted of  $\lg T$  on the  $y$ -axis and  $\lg l$  on the  $x$ -axis. Determine expressions for the gradient and  $y$ -intercept in terms of  $a$  and  $b$ .

gradient = .....

$y$ -intercept = .....

[1]

- (b) Values of  $l$  and  $t$  are given in Fig. 2.2.

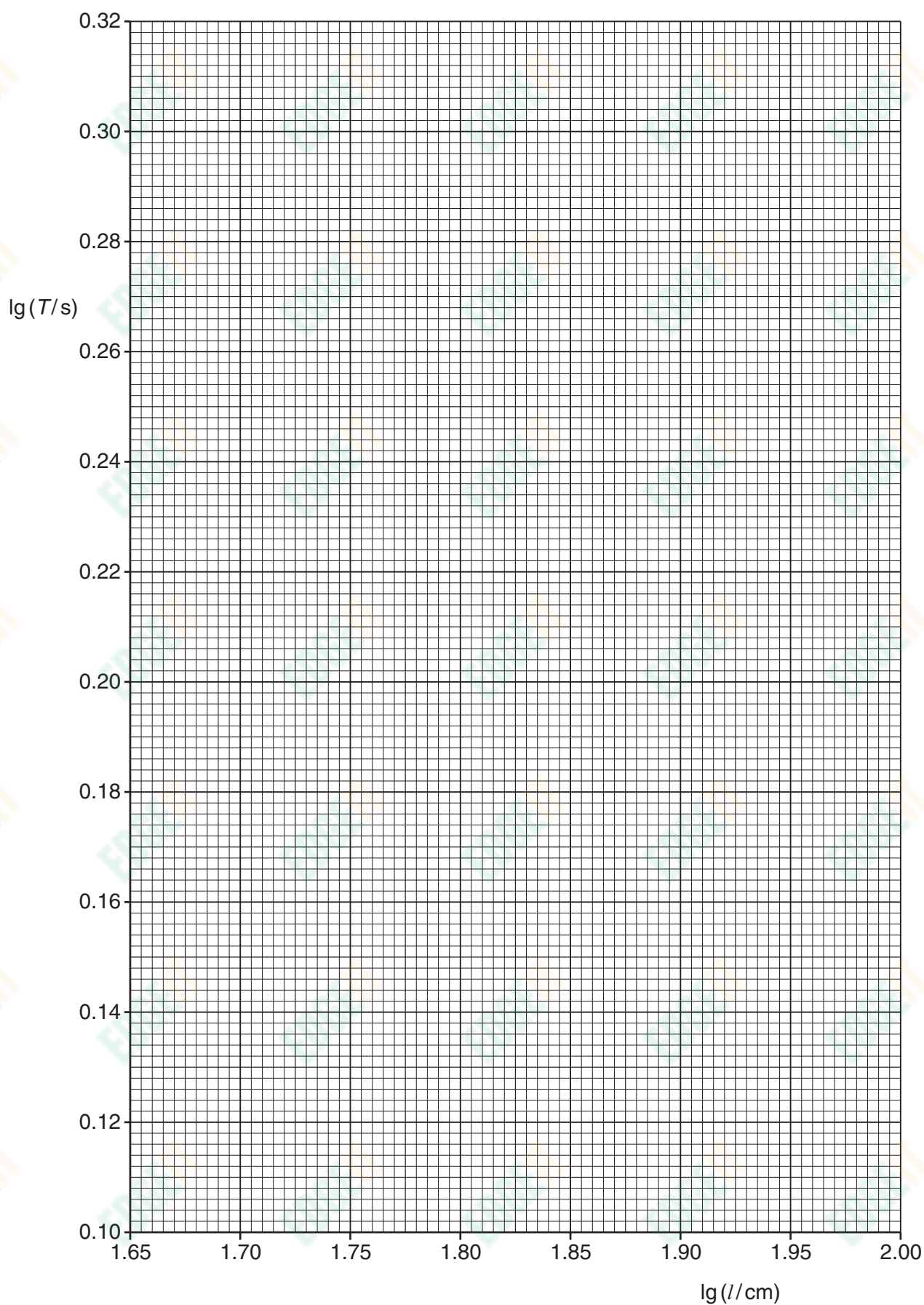
| $l / \text{cm}$ | $t / \text{s}$ | $T / \text{s}$ | $\lg (l / \text{cm})$ | $\lg (T / \text{s})$ |
|-----------------|----------------|----------------|-----------------------|----------------------|
| 95.0            | $19.6 \pm 0.2$ |                |                       |                      |
| 85.0            | $18.4 \pm 0.2$ |                |                       |                      |
| 75.0            | $17.4 \pm 0.2$ |                |                       |                      |
| 65.0            | $16.2 \pm 0.2$ |                |                       |                      |
| 55.0            | $14.8 \pm 0.2$ |                |                       |                      |
| 45.0            | $13.4 \pm 0.2$ |                |                       |                      |

Fig. 2.2

Calculate and record values of  $T / \text{s}$ ,  $\lg (l / \text{cm})$  and  $\lg (T / \text{s})$  in Fig. 2.2. Include the absolute uncertainties in  $\lg (T / \text{s})$ . [3]

- (c) (i) Plot a graph of  $\lg (T / \text{s})$  against  $\lg (l / \text{cm})$ . Include error bars for  $\lg (T / \text{s})$ . [2]
- (ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Both lines should be clearly labelled. [2]
- (iii) Determine the gradient of the line of best fit. Include the uncertainty in your answer.

gradient = ..... [2]



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- (iv) Determine the  $y$ -intercept of the line of best fit. Include the uncertainty in your answer.

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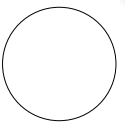
$y$ -intercept = ..... [2]

- (d) Using your answers to (c)(iii) and (c)(iv), determine values for  $a$  and  $b$ . Include the uncertainties in your answers. You need not be concerned with the units of  $a$  and  $b$ .

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$a$  = .....

$b$  = ..... [3]



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